Euclidean Geometry In Mathematical Olympiads

Problem 1.50 (IMO 2013/4)

IMO Chapter 1 geometry

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Kagebaka
3001 posts

Oct 18, 2018, 10:25 PM • 5 #1

Let ABC be an acute triangle with orthocenter H, and let W be a point on the side $\overline{BC}$, between B and C. The points M and N are the feet of the altitudes drawn from B and C, respectively. ω_1 is the circumcircle of triangle BWN and X is a point such that $\overline{WX}$ is a diameter of ω_1. Similarly, ω_2 is the circumcircle of triangle CWM and Y is a point such that $\overline{WY}$ is a diameter of ω_2. Show that the points X, Y, and H are collinear.

Kagebaka
3001 posts

Oct 18, 2018, 10:29 PM • 2 #2

In the book's solution it claims that $\angle XPW = 90$ (with P being the Miquel Point of M, N, W) somehow implies that X, P, H collinear?? I have no idea how this works 😕

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harry1234
1943 posts

Oct 24, 2018, 10:36 PM • 1 #4

We know that $\angle XPW = \angle APH = 90$, so HPW must also be 90, showing us that X, H, P are collinear.

This post has been edited 1 time. Last edited by harry1234, Oct 24, 2018, 10:38 PM

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illogical_21
1034 posts

Mar 17, 2019, 9:11 PM • 2 #5

Solution

> Let P be the miquel point of ABC, so $AMHPN, BWMP, CWNP$ cyclic, then $\measuredangle NPA = \measuredangle NHA = \measuredangle ABC$ and $\measuredangle NPW = \measuredangle NBW = \measuredangle ABC$ $\implies A, P, N$ collinear, now since $\measuredangle WPH = \measuredangle APH = \measuredangle AMH = 90^\circ = \measuredangle WPX$, we have P, H, X collinear, similarly, P, H, Y are collinear $\implies X, Y, H$ are collinear, as desired.

455264
74 posts

Oct 29, 2019, 2:38 PM • 1 #6

> *harry1234 wrote:*
> We know that $\angle XPW = \angle APH = 90$, so HPW must also be 90, showing us that X, H, P are collinear.

How do you know that A, P, H must be collinear?

PhysicsMonst...
1429 posts

Oct 29, 2019, 10:06 PM • 3 #7

> *Subh108 wrote:*
> How do you know that A, P, H must be collinear?

A, P and H are not collinear.

Hexagrammu...
1774 posts

Jan 5, 2020, 11:55 AM • 1 #8

> *Kagebaka wrote:*
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Yeah, this had me befuddled for suuuuuuch a long time! Only today I found out the correction on v_Enhance's blog.

DPS
812 posts

Jan 7, 2020, 12:37 PM • 1 #9

> *Hexagrammum16 wrote:*
>
> > *Kagebaka wrote:*
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